

Input metric

1. Input data $\mathbf{X}$

n points in $\mathbb{R}^m$
ambient coordinates



2. Build symmetric k -NN graph G_k

connect local neighbors
Euclidean edge weights



3. Compute $\Delta^{(k)}$ and Γ_k

all-pairs shortest paths
store canonical input geodesics



4. Initialize $\mathbf{Z}^{(0)}$

classical MDS on
 $\Delta^{(k)}$



Iterative geodesic SMACOF

5. Evaluate fixed geodesics

reuse stored $\gamma_{ij}^{(k)}$
compute d_k^{emb}



6. Accumulate path weights

$\tilde{\mathbf{v}}^{(t)}, \tilde{\mathbf{w}}^{(t)}$
from the fixed geodesic family



7. Update $\mathbf{Z}^{(t+1)}$

$(\tilde{\mathbf{w}}^{(t)} + \tilde{\mathbf{v}}^{(t)} \mathbf{Z}^{(t)})$
evaluate geodesic stress

if not converged

